

MEKORYUK, AK, USA

WMO: 702185

Lat: 60.372N

Lon: 166.272W

Elev: 45

StdP: 14.67

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26622

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.0	-9.3	-22.4	1.6	-10.8	-17.1	2.2	-9.2	40.9	23.3	37.4	24.9	8.9	30	0.962

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	61.0	54.8	57.4	52.6	55.4	51.4	55.9	59.8	53.9	57.1	52.3	55.4	12.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.1	62.2	58.0	52.1	57.9	56.4	50.1	53.7	54.4	23.7	59.6	22.5	57.6	21.5	55.5	66.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 36.0	32.4	28.3	DB	-18.4	66.4	4.0	5.0	-21.4	70.0	-23.7	72.9	-26.0	75.7	-28.9	79.3	(4)
(5)			WB	-18.7	58.6	3.6	3.0	-21.3	60.8	-23.4	62.6	-25.4	64.3	-28.0	66.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.0	12.0	15.7	15.4	25.0	35.6	43.6	49.0	50.5	45.9	34.9	25.8	17.0
	DBStd	16.06	13.20	12.07	11.56	8.73	5.42	4.59	3.49	3.69	4.31	6.15	8.48	11.81
	HDD50	7048	1178	961	1072	749	447	196	60	36	131	468	726	1024
	HDD65	12428	1643	1381	1537	1199	912	641	496	449	572	933	1176	1489
	CDD50	96	0	0	0	0	0	5	29	52	10	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	14.0	15.6	16.0	14.3	14.1	11.6	11.0	10.9	12.9	14.1	15.1	16.5	16.0
Precipitation	PrecAvg	28.1	1.9	1.9	1.3	1.2	1.5	2.1	3.0	3.9	3.8	2.8	2.6	2.1
	PrecMax	38.3	4.3	4.6	4.2	2.4	3.2	3.2	4.6	5.8	6.7	5.6	5.1	4.4
	PrecMin	22.6	0.4	0.6	0.2	0.3	0.2	0.9	1.8	1.3	0.6	0.8	0.5	0.5
	PrecStd	3.7	0.9	1.0	0.9	0.5	0.7	0.6	0.8	1.0	1.1	1.0	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	35.7	36.9	40.1	54.8	63.1	63.7	64.3	57.1	48.8	44.9	37.7
		MCWB	33.4	33.8	34.0	36.6	47.3	53.4	55.0	58.2	51.4	46.9	42.1	35.8
	2%	DB	34.3	34.3	34.7	37.2	49.3	56.8	60.6	61.2	55.0	46.4	41.0	36.1
		MCWB	32.3	32.7	32.8	33.9	44.5	50.0	54.2	55.9	50.8	44.0	39.0	34.3
	5%	DB	33.7	33.9	33.6	36.2	45.8	54.2	57.0	58.5	54.0	45.4	38.7	34.4
		MCWB	31.9	32.3	31.9	33.5	41.9	48.7	52.4	54.4	50.3	42.4	36.5	32.7
	10%	DB	30.3	32.1	31.5	34.5	43.4	51.8	54.9	56.0	52.3	43.4	36.5	33.5
		MCWB	29.4	31.2	30.8	32.4	39.7	47.4	51.2	52.4	49.0	39.8	34.5	31.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.4	34.2	34.6	37.8	48.0	54.3	57.0	59.2	53.8	47.7	42.2	36.8
		MCDB	34.4	34.5	36.0	40.3	54.1	61.1	61.4	63.3	55.3	48.6	44.4	38.0
	2%	WB	32.4	33.0	33.0	34.7	44.7	50.9	54.9	57.0	51.6	45.5	39.0	34.0
		MCDB	33.8	33.7	34.4	36.8	48.7	55.8	59.2	60.8	54.1	46.4	41.2	35.3
	5%	WB	31.7	32.1	31.9	33.4	42.2	49.0	53.1	55.0	50.5	42.7	37.2	32.7
		MCDB	33.3	33.3	33.1	35.2	45.8	53.4	56.5	58.2	53.1	45.2	38.7	34.1
	10%	WB	29.6	30.8	30.3	32.6	40.4	47.1	51.6	53.2	49.3	40.7	34.5	31.6
		MCDB	30.3	32.0	31.1	34.3	43.9	50.7	54.7	55.9	51.9	43.8	36.3	33.0
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.3	10.8	9.1	8.9	10.9	9.6	8.7	8.1	7.9	8.4	9.7
		MCDBR	9.0	9.1	7.8	8.1	13.8	17.0	13.9	12.4	9.6	9.6	8.3	8.2
		MCWBR	8.6	8.9	7.3	6.8	10.3	11.8	9.8	8.9	7.2	8.3	7.8	8.0
	5% WB	MCDBR	9.4	9.5	7.8	7.5	13.7	15.9	12.5	11.9	8.6	9.2	8.2	8.3
		MCWBR	9.2	9.4	7.4	6.5	10.3	11.3	9.0	8.9	6.9	8.4	7.9	8.2
Clear-Sky Solar Irradiance	taub	0.236	0.292	0.294	0.360	0.374	0.377	0.393	0.396	0.350	0.315	0.281	0.237	
	taud	2.068	2.110	2.186	2.045	2.228	2.381	2.366	2.332	2.424	2.333	2.066	1.928	
	Ebn at Noon	165	216	264	261	269	271	262	251	244	211	136	102	
	Edh at Noon	14	24	31	43	39	34	34	33	25	20	14	10	
All-Sky Solar Radiation	RadAvg	115	339	863	1341	1512	1600	1305	1015	670	347	113	51	
	RadStd	20	46	105	138	167	139	109	130	69	20	17	7	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page